

# Successive h-indices using OpenAlex

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## Abstract

I apply Schubert’s successive h-index framework to OpenAlex, computing  $h_2$  for 20,669 institutions and  $h_3$  for 214 countries across 17.1 million researchers. The overall institution-level ranking is, in substance, a ranking of medical school size: Medicine accounts for 44% of the author pool and structurally dominates any university-wide aggregate. Field-specific decomposition is more informative: Wageningen University leads Agricultural & Biological Sciences and Environmental Science while ranking 172nd overall; Carnegie Mellon leads Computer Science from 175th; Brazil holds two of the world’s six deepest dental-research benches. I normalize for group size using Egghe’s Lotkaian scaling, which accounts for 91–92% of variance in both  $h_2$  and  $h_3$ , and find that Italy and Japan reach nearly the same  $h_3$  from a  $6\times$  difference in institutional scale.

**Keywords:** successive h-index, OpenAlex, institutional ranking, research assessment

**JEL Classification:** I23 , C80

**MSC Classification:** 62P99 , 62-07 , 62P25

## 1 Introduction

In 2007, András Schubert published a paper titled “Successive h-indices.” (2007) It begins:

Hirsch’s excessively discussed h-index generated a long row of interpretations, applications and derivatives within a surprisingly short period. Symptomatically, Hirsch’s paper in the *Proceedings of the National Academy of Sciences of the United States of America* became the second highest cited

paper of the issue in which it had been published. In this note, attention is called to an apparently so far overlooked option: h-indices themselves may form the basis of a successive series of h-indices.

Given, e.g., the h-indices of the researchers of an institute (which indices will be denoted by  $h_1$  in what follows), an index,  $h_2$ , of the institute can be determined: the institute has an index  $h_2$  if  $h_2$  of its  $N$  researchers have an  $h_1$ -index of at least  $h_2$  each, and the other  $N - h_2$  researchers have  $h_1$ -indices lower than  $h_2$  each. The succession can then be continued, e.g., for networks of institutions or countries or other higher levels of aggregation.

Since no sufficient data for testing the concept on the researcher-institute-country hierarchy were available, another example was chosen.

Fifteen years later, OpenAlex (Priem, Piwowar, & Orr, 2022) launched a FOSS bibliographic catalogue complete with citation indexing, authors, fields, and institutions. Recent evaluations have found it to be a broadly suitable substitute for proprietary sources such as Scopus and Web of Science at the institutional level (Alperin, Portenoy, Demes, Larivière, & Haustein, 2024; Culbert et al., 2025), and I take it here as the dataset for which Schubert wished in 2007.

The idea of applying the h-index hierarchically at the researcher-institution level was proposed independently by Prathap (2006) in a one-page note less than a month before Schubert’s paper. Schubert’s distinctive contribution was to recognize that the construction extends to an arbitrary succession of levels, opening the path to  $h_3$  and beyond. In the years between, successive h-indices attracted modest but sustained attention. Egghe (2008) gave the concept a theoretical footing, showing how the exponent of the underlying Lotka distribution propagates through each successive aggregation step. Several case studies followed shortly after: Ruane and Tol (2008) computed a rational (non-integer) successive h-index across economics departments in Ireland; Arencibia-Jorge, Barrios-Almaguer, Fernández-Hernández, and Carvajal-Espino (2008) applied Schubert’s exact researcher-department-institution hierarchy to a Cuban research institute using Web of Science data; and Arencibia-Jorge and Rousseau (2009) applied Prathap’s framing to twelve Cuban brain-research institutions. None of this prior work, however, has had access to a comprehensive, open, institution-labeled snapshot of the kind OpenAlex now provides.

Schubert suggested that  $h_2$  of institutes would be more interesting than that of journals; Section 2 describes how I compute it at scale, and Section 3 presents the results.

## 2 Methodology

### 2.1 Data source and filtering

The underlying data is OpenAlex’s public authors snapshot (Priem et al., 2022), distributed as Parquet on `s3://openalex/data/parquet/authors/` (`us-east-1`, no credentials required) and partitioned into over 1,600 daily `updated_date=` directories totalling about 53 GB. Only four columns are read from each remote file (`id`, `h_index`, `affiliations`, `topics`), and an author is retained only if all of the following hold:

- `summary_stats.h_index` is non-null and greater than zero;
- at least one entry in `affiliations` has `institution.type = 'education'`;
- at least one entry in `topics` carries a non-null field classification.

Applying these filters to the full snapshot yields 17.1 million authors, spanning 20,669 institutions and OpenAlex’s current list of 26 fields. One limitation carried over from the source data: an author’s `h_index` is computed by OpenAlex from all works meeting its citation criteria regardless of retraction status, which OpenAlex records inconsistently (Hauschke & Nazarovets, 2025); a small, unquantified fraction of  $h_1$  values in the dataset may therefore be inflated by since-retracted work.

## 2.2 Author-level assignment

Each retained author is reduced to a single row: one institution and one field. An author may list several affiliations with `institution.type = 'education'`; rather than fractionally splitting or duplicating that author across institutions, I assign them to the one with which they were most recently affiliated. Recency is read from OpenAlex’s `affiliations` field, which records, for each institution an author has ever listed, the calendar years in which they published while there; I take the institution whose most recent such year is latest, breaking ties (identical latest year) by the numerically smallest OpenAlex institution identifier. I deliberately do not use the similarly-named `last_known_institutions` field for this: despite its name, it is simply the set of affiliations listed on an author’s single most recent work, carrying no per-entry recency of its own, so when an author has more than one such affiliation (e.g., a joint appointment or co-affiliated coauthors misattributed to one author) there is no way to tell which is “more recent” than another.

Field assignment works the same way but is weighted. OpenAlex tags each author with a set of topics, each carrying a `count` (roughly, the number of the author’s works classified under that topic) and a parent field. I sum `count` within each field and assign the author to whichever field carries the largest total weight: their modal field by publication volume, not simply their single most granular topic. The result is an `authors` table with one row per author: (`author_id`, `h_index`, `institution_id`, `institution_name`, `field`, `field_name`).

Two structural features of this procedure limit the impact of field classification errors. First, the paper operates at the *field* level (26 fields) rather than at the finer topic level ( $\sim 4,500$  topics); OpenAlex’s topic classifier fine-tunes a multilingual BERT model on title, abstract, and citation data and achieves a top-1 accuracy of 0.53 (OpenAlex, 2024), but many topic-level errors stay within the same parent field, so field-level accuracy is substantially higher. Second, an author’s field is determined by the mode of their *entire publication record*, not by a single work: a researcher would need a substantial fraction of their career output systematically misclassified to a different field before their assignment would change.

## 2.3 The h-index and its successive generalization

Hirsch’s (2005) original index, here  $h_1$ , is computed by OpenAlex per author from citation counts and taken as given. Schubert’s generalization (2007) lifts the same

definition up a level of aggregation: a group of  $N$  members, each already carrying an  $h_{k-1}$  index, has an index  $h_k = n$  if  $n$  of its members have  $h_{k-1} \geq n$  and the remaining  $N - n$  members have  $h_{k-1} < n$ :

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**Algorithm 1** Successive h-index of a group  $G$ , given a value  $v(m)$  for each member  $m \in G$

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- 1: Sort  $G$  by  $v$  descending; let  $r(m)$  be  $m$ 's 1-indexed rank in that order
  - 2: **return**  $\max\{r(m) : v(m) \geq r(m), m \in G\}$
- 

I use this rank-based procedure, implemented as a window function in SQL ranking rows within each group and taking the largest rank whose value has not yet fallen below it, for every successive index in this paper:  $h_2$  of an institution over the  $h_1$  of its authors, and  $h_3$  of a country over the  $h_2$  of its institutions.

## 2.4 $h_2$ : institution and institution $\times$ field

$h_2$  is computed twice, over different groupings of the same author-level data. The field-specific table groups authors by (institution, field) and yields 288,300 pairs; institutions are matched on their numeric identifier only; institution and field *names* are resolved afterward via `arg_max`, so that spelling or formatting variants of the same name across snapshot partitions cannot silently split one institution into several groups. The institution-overall table ignores field entirely and applies the same procedure across all of an institution's authors regardless of discipline, yielding one  $h_2$  per institution (20,669 rows).

## 2.5 $h_3$ : country level

Institutions are mapped to countries using the `country_code` field of OpenAlex's institutions snapshot (also read directly from S3, joined on institution ID).  $h_3$  is then computed with the same rank-based procedure, once globally per country over the  $h_2$  values of its institutions and once per (country, field) over the field-specific  $h_2$  values, producing one file per field.

## 2.6 Efficiency

Mryglod, Holovatch, and Kenna (2022) showed that group-level h-indices grow with group size even when per-member quality is held constant. They were not studying successive h indices, but a similar argument makes raw  $h_2$  and  $h_3$  values unsuitable for comparing institutions or countries of unequal scale.

Egghe (2008) applied a Lotkaian framework to successive h indices. Suppose the article-citation frequency function follows Lotka's law with exponent  $\alpha_0 > 1$ , and the author-article frequency function follows Lotka's law with exponent  $\alpha_1 > 1$ . Egghe showed that the distribution of  $h_1$  values across authors is itself Lotkaian with compound exponent  $\beta_1 = \alpha_0\alpha_1$ . Applying the h-index formula for Lotkaian systems, the institution-level index satisfies

$$h_2 = S^{1/\beta_1} \tag{1}$$

where  $S$  is the number of authors at the institution. An analogous argument at the next level gives the distribution of  $h_2$  values across institutions a Lotkaian exponent  $\beta_2 = \alpha_0\alpha_1\alpha_2$ , where  $\alpha_2$  is the Lotka exponent of the institution–author frequency function, so that

$$h_3 = R^{1/\beta_2} \quad (2)$$

where  $R$  is the number of institutions in the country. I define size-normalized efficiency metrics by dividing out these predicted scalings:

$$\varepsilon_2 = \frac{h_2}{S^{1/\beta_1}} \quad (3)$$

$$\varepsilon_3 = \frac{h_3}{R^{1/\beta_2}} \quad (4)$$

An institution with  $\varepsilon_2 > 1$  outperforms the Lotkaian prediction for its headcount; one with  $\varepsilon_2 < 1$  underperforms it.

The exponents  $\beta_1$  and  $\beta_2$  are estimated directly from the data rather than assembled from separately fit  $\alpha_0, \alpha_1, \alpha_2$ . I apply discrete maximum-likelihood tail estimation following [Clauset, Shalizi, and Newman \(2009\)](#), with  $x_{\min}$  chosen to minimize the Kolmogorov–Smirnov distance between the empirical and fitted CDFs, to the distributions of  $h_1$  across all authors and  $h_2$  across all institutions, reading  $\beta_1$  and  $\beta_2$  as their fitted Lotka exponents. I then fit  $\alpha_0, \alpha_1$ , and  $\alpha_2$  individually:  $\alpha_0$  from a stratified sample of per-article citation counts drawn from the OpenAlex works snapshot,  $\alpha_1$  from per-author work counts, and  $\alpha_2$  from per-institution author counts. Finally, I verify that  $\alpha_0\alpha_1 \approx \beta_1$  and  $\alpha_0\alpha_1\alpha_2 \approx \beta_2$  as a cross-check that the compounding-exponent structure of Egghe’s model holds in the OpenAlex dataset.

## 2.7 Supplementary metrics

Several secondary statistics are derived from the  $h_2$  and  $h_3$  tables to probe what the headline numbers obscure:

- **Gini coefficient**, computed on the within-field distribution of institutional  $h_2$ , quantifies whether a field’s excellence is concentrated in one institution (Gini near 1) or broadly distributed (Gini near 0).
- **Breadth score** counts, per institution, how many of the 26 fields it ranks in the global top 10, top 50, and top 100.
- **Non-biomedical ranking** recomputes institution-level  $h_2$  after excluding authors whose modal field is Medicine, Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, or Health Professions, to test whether the overall ranking is, in substance, a ranking of medical school size.

## 2.8 Manual cross-validation

Because errors in OpenAlex’s underlying author disambiguation or citation counts would propagate directly into every successive index, I manually spot-check a targeted, non-random sample of authors rather than a random one, chosen at the points where an error would be both most likely and most consequential: authors sitting

exactly at an institution’s  $h_2$  threshold, where a  $\pm 1$  error in a single  $h_1$  could change the institution’s rank; top authors in fields where OpenAlex’s topic classification is known to be thinner (e.g., Arts and Humanities); top authors at institutions with non-Anglophone naming conventions, where author-name disambiguation is more error-prone and where OpenAlex’s coverage is known to diverge unevenly from other databases by country, most severely for China (Zheng, Miao, Bu, & Larivière, 2025), including institutions suspected of affiliation-swapping for grant purposes (Bhattacharjee, 2011); and the small number of authors with the highest  $h_1$  globally, as a sanity check at the extreme of the distribution. For each candidate, I resolve the author’s OpenAlex profile through the public API and compare it by hand against Web of Science, Scopus, and Google Scholar. This design is deliberately adversarial: it is built to surface errors, not to estimate how frequently they occur across the full population, so any outlier rate found in it bounds the qualitative failure mode (which authors, and by how much) rather than the population-wide error rate.

## 2.9 Sensitivity to disambiguation error

Because the cross-validation sample above cannot itself bound a population-wide error rate, I separately ask how much author-disambiguation error the paper’s conclusions could tolerate before changing using a Monte Carlo sensitivity analysis run directly on the full authors table. At a population-wide error rate  $p$ , each author is independently flagged with probability  $p$ ; a flagged author’s  $h_1$  is multiplied by an inflation ratio drawn uniformly from the four ratios actually observed among the cross-validation outliers, simulating a merged-identity error of the same kind and magnitude found by hand. I recompute institution-level  $h_2$ , country-level  $h_3$ , and the field-specific  $h_2$  rankings, and compare each to the unperturbed baseline via Spearman rank correlation and top-set overlap. Rather than picking a single rate to call a “worst case,” I run five independent trials at each of six rates,  $p \in \{1\%, 5\%, 10\%, 20\%, 40\%, 80\%\}$ , tracing out how ranking stability responds to the assumed amount of error from a plausible rate up to a deliberately implausible one, so that the paper’s tolerance for this kind of error is characterized by a curve rather than asserted at a single point.

## 2.10 Robustness to field assignment

The single-field assignment described in Subsection 2.2 asks “which field accounts for the largest share of an author’s publication volume?” and assigns each author exclusively to that field. To test whether field-specific  $h_2$  rankings are stable under an alternative that allows multidisciplinary authors to count in multiple fields simultaneously, I re-run the field-specific  $h_2$  computation after expanding each author’s field membership: at a threshold  $\tau$ , an author is assigned to every field in which their OpenAlex topic-weight share is at least  $\tau$  (where topic-weight is the same count-weighted field sum used in the original assignment). I test two thresholds,  $\tau \in \{20\%, 33\%\}$ , spanning from liberal (an author need contribute only one fifth of their output to count in a field) to moderate (one third or more). For each threshold I recompute  $h_2$  for every (institution, field) pair in the expanded author roster, using the same rank-based algorithm as the original, and compare to the baseline via Spearman rank

correlation on shared (institution, field) pairs and by checking whether each field’s leading institution remains #1.

## 2.11 External validation against the Leiden Ranking

To assess whether  $\varepsilon_2$  captures genuine research quality rather than a residual size artefact, I compare both raw  $h_2$  and  $\varepsilon_2$  against the CWTS Leiden Ranking Open Edition 2024 (van Eck, 2024), an independent, OpenAlex-based institution-level bibliometric ranking based on mean normalised citation score (MNCS). I compute Spearman rank correlations  $\rho(h_2, \text{MNCS})$  and  $\rho(\varepsilon_2, \text{MNCS})$  on this matched set; p-values use the standard  $t$ -approximation  $t = \rho\sqrt{(n-2)/(1-\rho^2)}$  evaluated against a  $t_{n-2}$  distribution.

## 2.12 Software and reproducibility

All analyses were implemented in Python 3.10 (NumPy, pandas, geopandas, matplotlib, duckdb, powerlaw, scipy). The data fetching, filtering, and processing are fully reproducible from the public code repository at <https://github.com/rupumped/successive-h-indices>.

# 3 Results

## 3.1 Manual cross-validation

Of the 30 authors sampled under the scheme described in Subsection 2.8, Table 1 summarizes agreement between OpenAlex’s `h_index` and the corresponding value I located by hand in Web of Science, Scopus, and Google Scholar. Coverage itself varies sharply: Scopus returned a value for 26 of the 30 authors, Google Scholar for 19, and Web of Science for 16.

**Table 1** Agreement between OpenAlex and each external source, over authors for which that source returned a value. Diff is OpenAlex minus the external value.

| Source         | $n$   | Mean diff | Mean abs. diff | Worst case         |
|----------------|-------|-----------|----------------|--------------------|
| Web of Science | 16/30 | +24.5     | 25.7           | Lei Jiang (+100)   |
| Scopus         | 26/30 | +8.0      | 12.4           | Donald Small (+46) |
| Google Scholar | 19/30 | −22.7     | 30.7           | A. Jafari (−85)    |

Scopus tracks OpenAlex most closely, both in coverage and in magnitude of disagreement. Google Scholar is systematically higher than OpenAlex, consistent with its broader crawl of preprints, theses, and other grey literature and its comparative permissiveness toward self-citation; this ordering (Google Scholar > Scopus > OpenAlex > Web of Science) reproduces cleanly among the six global top- $h_1$  sanity-check authors, none of whom show a discrepancy exceeding 27% of their OpenAlex value in either direction.

Four authors, however, show disagreement well outside this range. I read this as a signature of author-disambiguation failure (multiple distinct people merged under one OpenAlex author ID) rather than of genuine cross-database disagreement:

- **Donald Small** (Johns Hopkins):  $h_1^{\text{OpenAlex}} = 103$  against Scopus = 57 and Google Scholar = 59.
- **Karen S. Calhoun** (University of Michigan):  $h_1^{\text{OpenAlex}} = 45$  against Scopus = 11 and Google Scholar = 13.
- **Lei Jiang** (University of Science and Technology of China):  $h_1^{\text{OpenAlex}} = 215$  against Web of Science = 115, consistent with documented China-specific coverage discrepancies in OpenAlex (Zheng et al., 2025).
- **A. Jafari** (Sharif University of Technology):  $h_1^{\text{OpenAlex}} = 139$ , but Web of Science, Scopus, and Google Scholar report 41, 133, and 224 respectively.

All four fall in the non-Anglophone-institution or common-name strata of the sample, matching the concern raised in Subsection 2.8 that disambiguation is more error-prone for exactly these authors. Because such merges can only inflate an author’s apparent  $h_1$ , and  $h_1$  feeds directly into  $h_2$  and  $h_3$ , a handful of merged identities at the right institution could measurably distort an institution- or country-level ranking; I flag this as an open limitation rather than one this sample can size definitively. This sample is deliberately adversarial, chosen to find errors, not to estimate how often they occur. The 4/30 (13%) outlier fraction should not be read as an estimate of how much of the full dataset is affected. The same pattern, disambiguation precision concentrated lower for common names and non-Anglophone institutions, appears in an independent benchmark of 2,700 scholars sampled systematically across three regions (Zhao & Chen, 2025), corroborating that this is a geographically and onomastically concentrated failure mode rather than an artifact of this particular sample. OpenAlex’s maintainers report an ongoing correction effort targeting exactly this kind of over-merging, with corresponding-author identification precision rising from 0.60 to 0.92 (F1 0.72 to 0.90) after a 2026 rework of the underlying author-matching model (Demes, 2026). Subsection 3.11 quantifies how much a population-wide error rate at or below the adversarial sample’s raw rate would actually move this paper’s rankings.

The four Arts and Humanities threshold authors sampled (Gillian Ramchand and G. Herbert Fowler at Oxford; D.H. Mellor and James Raven at Cambridge) illustrate the range of under-coverage across major citation databases: Ramchand and Fowler returned no value in any of the three external sources, while Mellor (OA  $h_1 = 28$ ) and Raven (OA  $h_1 = 24$ ) appeared in Scopus at roughly half their OpenAlex values ( $h_1 = 13$  and 12, respectively); for Raven, Google Scholar (26) brackets the OpenAlex value closely. This is consistent with the well-documented weaker coverage of book-based humanities scholarship in all major citation databases (Nederhof, 2006) and suggests that h-index comparisons in these fields should be read with correspondingly less confidence than in the sciences.

### 3.2 The overall $h_2$ ranking

Table 2 lists the ten institutions with the highest institution-level  $h_2$ , computed over all 20,932 institutions and 30.0 million authors described in Section 2.



**Table 2** Top 10 institutions by overall  $h_2$ .

| Institution                                 | $h_2$ | Authors |
|---------------------------------------------|-------|---------|
| Harvard University                          | 137   | 110,979 |
| Stanford University                         | 107   | 65,197  |
| University of Cambridge                     | 105   | 52,307  |
| Cornell University                          | 104   | 70,718  |
| Johns Hopkins University                    | 103   | 80,028  |
| University of Oxford                        | 103   | 57,110  |
| University of Washington                    | 102   | 71,270  |
| University of North Carolina at Chapel Hill | 101   | 64,478  |
| University College London                   | 100   | 77,482  |
| University of California, San Francisco     | 100   | 50,351  |

This broadly tracks the US News and World Report ranking, where Harvard also places 1st, Stanford 3rd, and Cambridge 5th. As I show below, however, the overall ranking is a considerably less interesting object than it first appears.

### 3.3 Field-level heterogeneity

Splitting  $h_2$  by (institution, field) reveals that the overall ranking is deceptive: several institutions that never come close to the top of Table 2 rank first globally in their field by this measure. Table 3 lists a sample of these cases.

University of Antwerp is the counterexample that proves the rule holds both ways: it is #14 overall ( $h_2 = 98$ ) *and* #1 in Physics and Astronomy ( $h_2 = 96$ ), a broad powerhouse that also ranks first in a specific, non-biomedical field. Universidade Estadual de Campinas (UNICAMP) places 6th in Dentistry behind São Paulo, placing Brazil among the top six globally in Dentistry by this measure despite the country appearing nowhere near the top of any general-purpose ranking (Vaccaro, Sánchez-Núñez, & Witt-Rodríguez, 2022). China’s cluster is not one specialty but four related applied-science fields: University of Science and Technology of China (USTC) ranks first in Materials Science, Engineering, and Energy by field  $h_2$ , while its sister institution, the University of Chinese Academy of Sciences, ranks first in Chemical Engineering.

Not every field supports this kind of comparison equally well. Decision Sciences is the thinnest field in the dataset: Stanford’s global #1 is  $h_2 = 22$ , built on only 285 Stanford authors against 140,147 authors field-wide, versus Medicine’s  $h_2 = 119$  built on 7.58 million authors field-wide.  $h_2$  is not meaningfully comparable across fields of such different size, only within a field, and Decision Sciences in particular may be too sparse an OpenAlex topic category for its ranking to carry much signal.

### 3.4 Robustness of field-specific rankings to multi-field assignment

The single-field assignment described in Subsection 2.2 asks which field accounts for the plurality of an author’s publication volume, and assigns them exclusively there. To test how much this choice moves the field-level rankings, I re-run the field-specific

**Table 3** Selected field leaders that rank well outside the overall top 10. “Overall” columns give the institution’s rank and  $h_2$  in Table 2’s full ranking; “Field  $h_2$ ” is the institution’s  $h_2$  restricted to that field.

| Field                                    | Leader                               | Ov. rank | Ov. $h_2$ | Field $h_2$ |
|------------------------------------------|--------------------------------------|----------|-----------|-------------|
| Agricultural & Biological Sciences       | Wageningen                           | 172      | 71        | 56          |
| Environmental Science                    | Wageningen                           | 172      | 71        | 56          |
| Pharmacology, Toxicology & Pharmaceutics | Shenyang Pharmaceutical University   | 535      | 53        | 24          |
| Computer Science                         | Carnegie Mellon University           | 175      | 70        | 56          |
| Materials Science                        | USTC                                 | 39       | 89        | 59          |
| Engineering                              | USTC                                 | 39       | 89        | 62          |
| Energy                                   | USTC                                 | 39       | 89        | 52          |
| Chemical Engineering                     | Univ. of Chinese Academy of Sciences | 91       | 80        | 23          |
| Dentistry                                | Universidade de São Paulo            | 235      | 66        | 34          |

$h_2$  computation after allowing each author to count in every field for which their OpenAlex topic-weight share is at least  $\tau$  in accordance with Subsection 2.10.

At  $\tau = 33\%$  (an author must contribute at least one third of their output to count in a field), the (institution, field) pair count rises by 9.3% (333,132  $\rightarrow$  364,032 pairs). The Spearman rank correlation between original and multi-field  $h_2$  values, computed on the inner join of shared pairs, is  $\rho = 0.963$ : field rankings are highly stable. Of the 27 fields’ leading institutions, 23 are unchanged. The Wageningen Ag&Bio leadership (Table 3) survives and Carnegie Mellon’s Computer Science lead survives. The one notable exception is Environmental Science: the University of Colorado Boulder overtakes Wageningen at this threshold, because Boulder has a larger body of researchers who do substantial but not modal Environmental Science work. The two methods answer different questions. The single-field assignment identifies institutions where Environmental Science is researchers’ *primary* field; the multi-field assignment identifies institutions with the most researchers for whom it is a *substantial* field.

At  $\tau = 20\%$  (a more aggressive threshold admitting all authors with one fifth or more of their output in a field), the pair count rises by 20.7% and  $\rho = 0.907$ ; 18 of 27 field leaders survive. Carnegie Mellon’s Computer Science lead also falls at this threshold, with Stanford leading instead, consistent with the same mechanism of Stanford having a larger population of researchers with at least 20% CS output, even if CS is not their modal field. The Wageningen Ag&Bio and Environmental Science results are unchanged from the 33% case. These findings confirm that the

overall field ranking structure is broadly stable under multi-field assignment but that the field-leadership claims for smaller, highly specialised institutions depend on how narrowly “counts in this field” is defined. Given this sensitivity, the field-leadership claims in Table 3 should be read as reflecting depth of specialisation under the paper’s single-field assignment.

### 3.5 Concentration within fields

Table 4 reports, for every field, the Gini coefficient of institutional  $h_2$  defined in Section 2.7, together with the field’s institution count, leading (maximum)  $h_2$ , and mean  $h_2$ .

**Table 4** Gini coefficient of institutional  $h_2$  within each field, sorted from most to least concentrated.

| Field                                      | Institutions | Max $h_2$ | Mean $h_2$ | Gini  |
|--------------------------------------------|--------------|-----------|------------|-------|
| Physics and Astronomy                      | 12,739       | 96        | 6.11       | 0.605 |
| Biochemistry, Genetics & Molecular Biology | 15,404       | 104       | 6.61       | 0.578 |
| Neuroscience                               | 11,646       | 64        | 4.41       | 0.574 |
| Immunology and Microbiology                | 9,838        | 67        | 3.97       | 0.561 |
| Materials Science                          | 13,097       | 59        | 6.36       | 0.560 |
| Earth and Planetary Sciences               | 12,447       | 60        | 4.30       | 0.557 |
| Chemistry                                  | 12,103       | 41        | 5.02       | 0.541 |
| Environmental Science                      | 16,603       | 56        | 5.54       | 0.541 |
| Engineering                                | 17,742       | 62        | 6.91       | 0.533 |
| Medicine                                   | 19,119       | 119       | 8.58       | 0.528 |
| Agricultural and Biological Sciences       | 16,391       | 56        | 5.61       | 0.522 |
| Energy                                     | 8,971        | 52        | 3.38       | 0.515 |
| Psychology                                 | 15,745       | 58        | 3.64       | 0.513 |
| Computer Science                           | 16,736       | 56        | 5.08       | 0.512 |
| Dentistry                                  | 8,303        | 34        | 3.01       | 0.500 |
| Mathematics                                | 11,092       | 36        | 3.42       | 0.491 |
| Economics, Econometrics and Finance        | 13,923       | 55        | 3.30       | 0.482 |
| Business, Management and Accounting        | 14,623       | 32        | 3.59       | 0.481 |
| Health Professions                         | 14,840       | 41        | 2.88       | 0.465 |
| Social Sciences                            | 19,242       | 48        | 4.57       | 0.461 |
| Pharmacology, Toxicology and Pharmaceutics | 7,219        | 24        | 2.61       | 0.459 |
| Chemical Engineering                       | 5,874        | 23        | 2.48       | 0.446 |
| Arts and Humanities                        | 15,390       | 30        | 2.78       | 0.443 |
| Nursing                                    | 8,958        | 31        | 2.37       | 0.433 |
| Decision Sciences                          | 10,409       | 22        | 2.37       | 0.423 |
| Veterinary                                 | 4,678        | 28        | 2.04       | 0.422 |

Physics and Astronomy is the most unequal field: University of Antwerp’s  $h_2 = 96$  is 15.7 times the field mean of 6.1. Veterinary and Decision Sciences are the most evenly distributed (Gini  $\approx 0.42$ ). Medicine sits in the middle of the table (0.528) despite

Harvard’s large absolute lead, because Medicine, unlike Physics and Astronomy, has thousands of institutions with meaningful  $h_2$  rather than a single dominant outlier.

### 3.6 The overall ranking as an implicit biomedical ranking

Medicine alone accounts for 7.58 million of the dataset’s 30.0 million authors (25%), and, together with Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, and Health Professions, the five core biomedical fields dwarf every other field’s author pool. Table 5 recomputes institution-level  $h_2$  after excluding authors in these five fields entirely to test whether the overall ranking in Table 2 is, in substance, a ranking of medical school size.

**Table 5** Top 10 institutions by non-biomedical  $h_2$ , alongside their position in the unrestricted overall ranking.  $\Delta$ rank is overall rank minus non-biomedical rank; positive means the institution moves up once biomedical fields are excluded.

| Institution                              | Rank | $h_2$ | Overall rank | Overall $h_2$ | $\Delta$ rank |
|------------------------------------------|------|-------|--------------|---------------|---------------|
| Harvard University                       | 1    | 100   | 1            | 137           | 0             |
| University of Antwerp                    | 2    | 97    | 14           | 98            | +12           |
| California Institute of Technology       | 3    | 94    | 15           | 97            | +12           |
| Stanford University                      | 4    | 92    | 2            | 107           | −2            |
| Université Paris-Saclay                  | 5    | 91    | 22           | 95            | +17           |
| Massachusetts Institute of Technology    | 6    | 89    | 25           | 94            | +19           |
| University of Geneva                     | 7    | 88    | 29           | 93            | +22           |
| Univ. of Science and Technology of China | 8    | 88    | 39           | 89            | +31           |
| ETH Zurich                               | 9    | 87    | 36           | 89            | +27           |
| University of Cambridge                  | 10   | 87    | 3            | 105           | −7            |

Harvard remains #1 even with its biomedical authors removed, confirming the claim in Subsection 3.3 that it is genuinely excellent rather than merely large. But the rest of the table reshuffles substantially: Antwerp, Caltech, Paris-Saclay, MIT, Geneva, USTC, and ETH Zurich all gain 12 to 31 places once Medicine and its four sibling fields are set aside, while Stanford and Cambridge, both anchored by large medical schools, fall by 2 and 7 places respectively. The University of Chinese Academy of Sciences shows the same pattern one tier down: 91st overall but 23rd non-biomedical. Because medical fields have by far the largest, most-cited author pools, they structurally dominate any university-wide aggregate; an institution elite in non-medical fields, such as USTC or Carnegie Mellon, can never reach the top of the overall list no matter how good it is, simply because it lacks a large medical school. The field-level and non-biomedical breakdowns matter more than the unrestricted university-level number.

### 3.7 Breadth versus depth

Table 3 showed institutions that lead a single field while sitting far outside the overall top 100. Table 6 shows the opposite pattern: institutions whose strength is broad

rather than deep, measured by how many of the 26 fields they place in the global top 50.

**Table 6** The nine institutions with the most top-50 field placements out of 26 fields.

| Institution                            | Top 50 | Top 10 | Overall rank | Overall $h_2$ |
|----------------------------------------|--------|--------|--------------|---------------|
| University of California, Los Angeles  | 20     | 8      | 13           | 99            |
| University of Michigan                 | 20     | 7      | 26           | 94            |
| Harvard University                     | 19     | 13     | 1            | 137           |
| University of Oxford                   | 19     | 8      | 6            | 103           |
| University of Washington               | 17     | 8      | 7            | 102           |
| University of Toronto                  | 17     | 7      | 11           | 100           |
| Cornell University                     | 17     | 6      | 4            | 104           |
| Univ. of North Carolina at Chapel Hill | 17     | 5      | 8            | 101           |
| University of Cambridge                | 17     | 4      | 3            | 105           |

UCLA and the University of Michigan share the greatest breadth by this measure, each placing in the global top 50 in 20 of 26 fields; UCLA leads 8 outright while Michigan leads 7. Harvard still leads more fields outright (13 of 26) than any other institution, but four of the nine broadest institutions by this top-50 count, Toronto, UNC Chapel Hill, Washington, and Cambridge, place fewer than half their 17 top-50 fields inside the top 10, meaning most of their breadth comes from being solidly good across many fields rather than dominant in any one. At the other extreme, Boston University (rank 23,  $h_2 = 94$ ) achieves its overall strength with zero top-10 field placements, through scale and consistency across many merely-good departments rather than fielding the single best group in any one discipline.

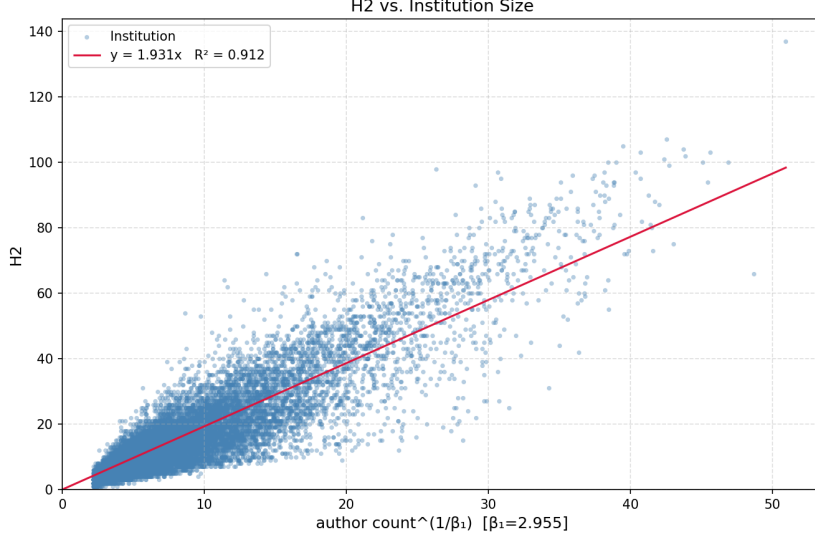
### 3.8 Institution efficiency

#### Model validation

Figure 1 plots institution-level  $h_2$  against the Egghe-predicted value  $S^{1/\beta_1}$  for all institutions with at least 10 OpenAlex authors. A no-intercept regression, appropriate because  $N = 0$  authors forces  $h_2 = 0$ , gives slope  $c = 1.945$  and uncentered  $R^2 = 0.911$ :  $S^{1/\beta_1}$  alone explains 91% of the variance in  $h_2$ , confirming that Egghe’s power-law functional form describes the data. The efficiency metric  $\varepsilon_2 = h_2/S^{1/\beta_1}$  captures each institution’s deviation from this overall trend.

The cross-check on Egghe’s compounding structure fails sharply, however. Egghe’s derivation requires that the Lotka tail exponent of the  $h_1$  distribution equals the product  $\alpha_0\alpha_1$ ; these separate fits give  $\alpha_0 = 2.742$  and  $\alpha_1 = 2.996$ , whose product is 8.21, against the directly-measured  $\beta_1 = 2.963$ : a discrepancy of 177%. The analogous prediction at the next level ( $\beta_2 = \beta_1\alpha_2$ ) is similarly violated:  $2.963 \times 1.994 = 5.909$  against the directly-measured  $\beta_2 = 2.998$  (97% discrepancy).

The failure of the compounding formula does not, however, undermine the efficiency metric. The two claims play different roles. The compounding formula ( $\beta_1 =$



**Fig. 1** Institution-level  $h_2$  versus  $S^{1/\beta_1}$  (Egghe’s predicted scaling,  $\beta_1 = 2.963$ ). No-intercept regression:  $h_2 = 1.945 S^{1/\beta_1}$ ,  $R^2 = 0.911$  (uncentered).

$\alpha_0\alpha_1$ ) is a *theoretical prediction* about the relationship between separately-fitted exponents; it is tested here and found wanting. The power-law scaling  $h_2 \propto S^{1/\beta_1}$  is an *empirical finding* ( $R^2 = 0.911$ ) that does not depend on the compounding formula being correct.  $\beta_1$  is estimated directly from the distribution of  $h_1$  values across authors, not derived from  $\alpha_0\alpha_1$ , so the 177% discrepancy in the compounding formula has no bearing on the exponent actually used in  $\varepsilon_2$ . This can be externally validated against the Leiden Ranking later in this subsection to provide an independent, data-driven confirmation based on the correlation between  $\varepsilon_2$  and mean normalised citation score.

### Efficiency rankings

The efficiency metric  $\varepsilon_2 = h_2/S^{1/\beta_1}$  addresses Mryglod et al.’s (2022) concern directly: because  $S^{1/\beta_1}$  accounts for 91% of the variance in  $h_2$ , dividing it out isolates the residual variation that is not explained by headcount alone.

Table 7 lists the top 10 institutions by  $\varepsilon_2$  among the 12,026 with at least 100 authors. The top of the ranking is dominated by small and highly specialized institutions. Focused research centers and specialized colleges recur throughout the top 20.

Table 8 restricts to the 1,339 institutions with at least 5,000 authors. The leading institutions are now a mix of specialized research institutes like Rockefeller University ( $\varepsilon_2 = 3.77$ , biomedical), the Institute of Cancer Research ( $\varepsilon_2 = 3.57$ ), and the London School of Hygiene & Tropical Medicine ( $\varepsilon_2 = 3.44$ ) and comprehensive universities like the University of Antwerp ( $\varepsilon_2 = 3.72$ ,  $h_2 = 98$ , efficiency rank 112). The University of California System, treated here as a single institution under its OpenAlex identifier, is the only entry in the top 100 on both metrics simultaneously ( $h_2$  rank 73, efficiency

**Table 7** Top 10 institutions by efficiency  $\varepsilon_2 = h_2/S^{1/\beta_1}$  ( $\beta_1 = 2.963$ ) among those with at least 100 authors ( $n = 12,026$ ).

| Institution                                        | $\varepsilon_2$ | $h_2$ | Authors | $h_2$ rank |
|----------------------------------------------------|-----------------|-------|---------|------------|
| Bellevue University                                | 6.24            | 54    | 590     | 497        |
| Augustana University                               | 5.61            | 64    | 1,332   | 264        |
| Center for Behavioral Brain Sciences               | 5.48            | 31    | 168     | 1,908      |
| Barcelona Inst. of Science and Technology          | 5.43            | 53    | 839     | 520        |
| Imperial Valley College                            | 5.33            | 62    | 1,412   | 319        |
| Saint Camillus International Univ. of Health Sci.  | 5.30            | 39    | 364     | 1,228      |
| City College of San Francisco                      | 4.92            | 43    | 605     | 934        |
| Frederick Community College                        | 4.81            | 37    | 416     | 1,331      |
| Académie Nationale de Médecine                     | 4.72            | 39    | 512     | 1,192      |
| Natl. Interuniversity Consortium of Materials Sci. | 4.67            | 40    | 571     | 1,150      |

rank 67). Among the comprehensive elite research universities that dominate the raw  $h_2$  table, Caltech (efficiency rank 367), the University of Geneva (344), and Université Paris-Saclay (465) are the most efficient, each achieving  $h_2 \geq 93$  with under 25,000 authors.

**Table 8** Top 10 institutions by efficiency  $\varepsilon_2 = h_2/S^{1/\beta_1}$  ( $\beta_1 = 2.963$ ) among those with at least 5,000 authors ( $n = 1,339$ ).

| Institution                                  | $\varepsilon_2$ | $h_2$ | Authors | $h_2$ rank |
|----------------------------------------------|-----------------|-------|---------|------------|
| University of California System              | 3.93            | 83    | 8,242   | 73         |
| Rockefeller University                       | 3.77            | 70    | 5,637   | 179        |
| University of Antwerp                        | 3.72            | 98    | 15,759  | 14         |
| Institute of Cancer Research                 | 3.57            | 66    | 5,570   | 230        |
| University of San Diego                      | 3.56            | 64    | 5,097   | 286        |
| California University of Pennsylvania        | 3.49            | 76    | 9,011   | 121        |
| University of Wuppertal                      | 3.48            | 68    | 6,515   | 209        |
| London School of Hygiene & Tropical Medicine | 3.44            | 65    | 5,903   | 249        |
| Royal Holloway, University of London         | 3.39            | 61    | 5,102   | 339        |
| Vita-Salute San Raffaele University          | 3.37            | 71    | 8,171   | 171        |

## External validation against the Leiden Ranking

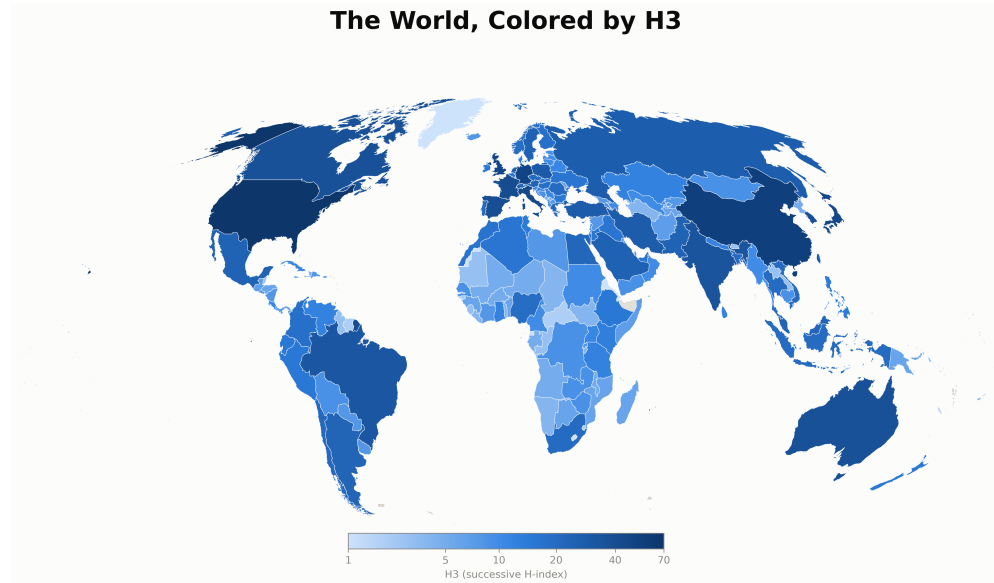
Matching the 1,506 Leiden Ranking Open Edition 2024 universities to OpenAlex’s dataset yields 1,495 institutions with both an MNCS value and an institution-wide  $h_2$  from OpenAlex. The Spearman rank correlation between raw  $h_2$  and MNCS is  $\rho = 0.634$  ( $p \approx 6 \times 10^{-170}$ ); between  $\varepsilon_2$  and MNCS it is  $\rho = 0.580$  ( $p \approx 5 \times 10^{-135}$ ). The successive h-index hierarchy captures genuine differences in research quality across institutions rather than differences in size alone.

### 3.9 $h_3$ : country level

Aggregating institutional  $h_2$  up to the country level via the same rank-based procedure yields  $h_3$ . Table 9 lists the ten highest- $h_3$  countries, and Figure 2 maps  $h_3$  for all 214 countries with at least one qualifying institution.

**Table 9** Top 10 countries by  $h_3$ .

| Country        | $h_3$ | Institutions |
|----------------|-------|--------------|
| United States  | 70    | 4,242        |
| China          | 55    | 1,078        |
| Germany        | 49    | 505          |
| United Kingdom | 47    | 548          |
| Japan          | 46    | 1,358        |
| Italy          | 45    | 229          |
| France         | 41    | 394          |
| Spain          | 39    | 156          |
| South Korea    | 37    | 358          |
| Australia      | 36    | 120          |



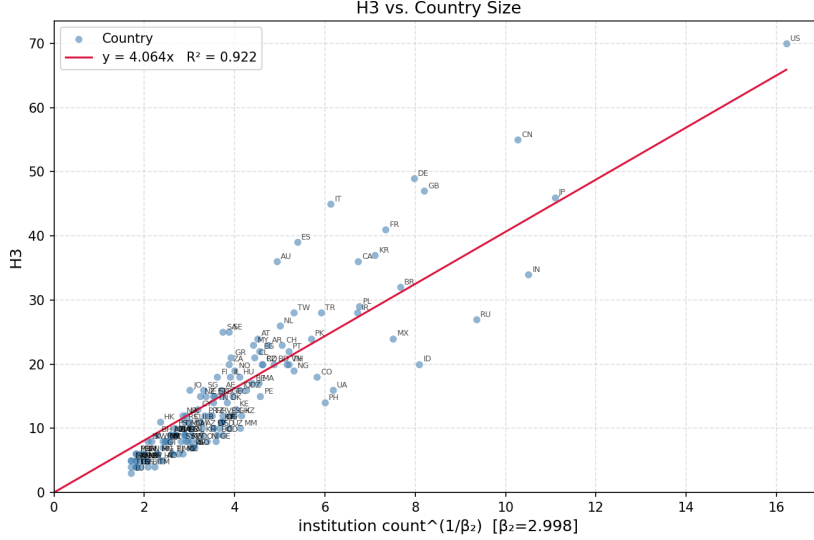
**Fig. 2**  $h_3$  by country. Countries with no qualifying institution are shown in grey.

The United States leads by a wide margin, but Italy's 45 on only 229 institutions and Japan's 46 on 1,358 are a first hint that raw  $h_3$  rewards national research-system size as well as depth, a point to which I return in Subsection 3.10.



### 3.10 Country efficiency

Figure 3 repeats the size-scaling analysis at the country level, plotting  $h_3$  against  $R^{1/\beta_2}$  for each country. The no-intercept fit gives slope  $c = 4.064$  and uncentered  $R^2 = 0.922$ : institution count alone explains 92% of the variance in  $h_3$  across countries.



**Fig. 3** Country-level  $h_3$  versus  $R^{1/\beta_2}$  (Egghe’s predicted scaling,  $\beta_2 = 2.998$ ). No-intercept regression:  $h_3 = 4.064 R^{1/\beta_2}$ ,  $R^2 = 0.922$  (uncentered).

Table 10 ranks the 151 countries with at least 5 qualifying institutions by efficiency  $\varepsilon_3 = h_3/R^{1/\beta_2}$ . The Italy–Japan contrast foreshadowed in Subsection 3.9 is now quantified: Italy ( $\varepsilon_3 = 7.35$ , rank 1) achieves  $h_3 = 45$  from only 229 institutions, while Japan ( $\varepsilon_3 = 4.15$ , rank 44) reaches  $h_3 = 46$  from 1,358, virtually the same headline number but at  $6\times$  the institutional scale. Australia (rank 2,  $\varepsilon_3 = 7.29$ , 120 institutions) and Spain (rank 3,  $\varepsilon_3 = 7.24$ , 156 institutions) round out a top tier of countries with compact but high-quality research systems. Germany (rank 6), the United Kingdom (rank 7), and France (rank 8) are the only countries that place in the top 10 on both raw  $h_3$  and efficiency, confirming that their research depth is genuine rather than an artefact of institutional scale. Saudi Arabia is a notable outlier at 4th by efficiency ( $h_3=25$  on just 52 institutions) despite ranking only 20th by raw  $h_3$ ; as flagged in Subsection 2.8, this is plausibly an artifact of high-h-index researchers listing Saudi institutions as secondary affiliations in exchange for funding (Bhattacharjee, 2011) rather than a genuine concentration of resident research talent.

The United States falls to efficiency rank 34 ( $\varepsilon_3 = 4.32$ ) despite its raw lead ( $h_3 = 70$ , rank 1). With 4,242 qualifying institutions, nearly  $8\times$  more than China and  $18\times$  more than Italy, even a large absolute  $h_3$  advantage is absorbed by the size denominator. China (rank 10,  $\varepsilon_3 = 5.36$ , 1,078 institutions) is more compact and appears above the US on this metric.

**Table 10** Top 10 countries by efficiency  
 $\varepsilon_3 = h_3/R^{1/\beta_2}$  ( $\beta_2 = 2.998$ ) among those with at least  
5 institutions ( $n = 151$ ).

| Country        | $\varepsilon_3$ | $h_3$ | Institutions | $h_3$ rank |
|----------------|-----------------|-------|--------------|------------|
| Italy          | 7.35            | 45    | 229          | 6          |
| Australia      | 7.29            | 36    | 120          | 10         |
| Spain          | 7.24            | 39    | 156          | 8          |
| Saudi Arabia   | 6.69            | 25    | 52           | 20         |
| Sweden         | 6.45            | 25    | 58           | 21         |
| Germany        | 6.14            | 49    | 505          | 3          |
| United Kingdom | 5.74            | 47    | 548          | 4          |
| France         | 5.59            | 41    | 394          | 7          |
| Greece         | 5.36            | 21    | 60           | 31         |
| China          | 5.36            | 55    | 1,078        | 2          |

### 3.11 Sensitivity to disambiguation error

Table 11 reports, for six simulated population-wide error rates spanning  $p = 1\%$  to a deliberately implausible  $p = 80\%$  (Subsection 2.9), the Spearman rank correlation and top-set overlap between the perturbed and unperturbed rankings, averaged over five trials, together with how often the paper’s two headline field-leadership claims survive the perturbation intact. The cross-validation sample’s own raw outlier fraction ( $4/30 \approx 13\%$  across both waves, Subsection 3.1) is included as an extra row for reference, though it carries no special status in this analysis.

Institution- and country-level rank correlations decline only modestly with  $p$  and then plateau rather than continuing toward zero: Spearman  $\rho(h_2)$  falls from 0.999 at  $p = 1\%$  to 0.991 at  $p = 20\%$  and then stays flat at 0.988 all the way out to  $p = 80\%$ ;  $\rho(h_3)$  shows the same pattern, from 0.9997 down to 0.994. This follows from how  $h_2$  and  $h_3$  are computed: each is an order statistic (the largest rank whose value has not yet fallen below it), so what matters for a given institution or country is whether a noisy author sits near *its own* threshold, not how many noisy authors exist elsewhere in the dataset; and because the perturbation is bounded in magnitude (at most  $\approx 4\times$ , the largest of the four observed ratios) and applied independently and identically everywhere, raising  $p$  further increasingly inflates competing institutions’ threshold authors by similar amounts, which cancels out much of the relative shift.

The field-specific leadership claims for Wageningen (Agricultural & Biological Sciences, Environmental Science) and Carnegie Mellon (Computer Science) are noisier with five trials per rate: each survives in 3 to 5 of 5 trials at every rate tested. Wageningen’s Environmental Science and Carnegie Mellon’s Computer Science leads are each knocked out in exactly two trials across the entire grid; Wageningen’s Agricultural & Biological Sciences lead is the most sensitive, failing in four trials (one at  $p = 10\%$ , one at  $p = 13\%$ , and two at  $p = 20\%$ ). Italy and Japan’s near-parity in  $h_3$  despite a  $6\times$  difference in institutional scale is preserved throughout the tested range: both values rise together under added noise (Italy 45 at  $p = 1\%$  to 66–67 at  $p = 80\%$ ; Japan 46 to 86–88 over the same range), and while their ratio drifts from about 0.96 down

to about 0.76 as  $p$  increases, even at that extreme it remains more than four times tighter than the underlying 1:6 gap in institutional count.

Reading these together: the paper’s overall institution and country rankings, and its country-comparison claims, are robust to author-disambiguation error at every rate tested, including rates far beyond what any plausible reading of the cross-validation sample would suggest. The field-leadership claims are more sensitive, as single-institution results necessarily are, and should be read with correspondingly more caution than the paper’s other claims.

**Table 11** Sensitivity of  $h_2$  and  $h_3$  rankings to simulated disambiguation error, by population-wide error rate  $p$  (5 trials per rate). Overlap is computed on the top 20 institutions (by  $h_2$ ) and top 10 countries (by  $h_3$ ). The last three columns give the fraction of trials in which the named institution keeps rank #1 in the stated field.

| $p$  | $\rho(h_2)$ | Top-20 | $\rho(h_3)$ | Top-10 | Wag. Ag&Bio #1 | Wag. EnvSci #1 | CMU CS #1 |
|------|-------------|--------|-------------|--------|----------------|----------------|-----------|
| 1%   | 0.999       | 93%    | 0.9997      | 94%    | 5/5            | 5/5            | 5/5       |
| 5%   | 0.997       | 84%    | 0.999       | 96%    | 5/5            | 5/5            | 5/5       |
| 10%  | 0.994       | 78%    | 0.998       | 90%    | 4/5            | 5/5            | 5/5       |
| 13%* | 0.993       | 78%    | 0.998       | 90%    | 4/5            | 5/5            | 5/5       |
| 20%  | 0.991       | 80%    | 0.997       | 90%    | 3/5            | 4/5            | 3/5       |
| 40%  | 0.988       | 79%    | 0.996       | 90%    | 5/5            | 4/5            | 5/5       |
| 80%  | 0.988       | 83%    | 0.994       | 90%    | 5/5            | 5/5            | 5/5       |

\*cross-validation sample’s raw outlier fraction, included for reference only.

## 4 Conclusion

Schubert closed his 2007 note wondering whether  $h_2$  of institutes would prove more interesting than  $h_2$  of journals, absent the researcher–institute–country data needed to check. Nineteen years and 30.0 million OpenAlex authors later, the answer is yes, but not in the way a single top-10 table would suggest.

The overall  $h_2$  ranking is the least interesting object in this paper. Because Medicine alone accounts for 25% of the author pool, the unrestricted university-wide aggregate is, in substance, a ranking of medical school size. The field decomposition reveals a completely different map of global research strength: Wageningen ranks first in Agricultural & Biological Sciences and Environmental Science by field  $h_2$  while ranking 172nd overall, though the Environmental Science lead holds only under the paper’s single-field assignment and not under multi-field assignment (Subsection 3.4); Carnegie Mellon ranks first in Computer Science by field  $h_2$  from 175th; the University of Science and Technology of China ranks first in Materials Science, Engineering, and Energy from 39th; and Shenyang Pharmaceutical University ranks first in Pharmacology from 535th. Brazil offers perhaps the sharpest illustration of what the overall ranking hides: the Universidade de São Paulo and the Universidade Estadual de Campinas rank among the top six globally in Dentistry by field  $h_2$  despite neither country nor either institution appearing anywhere near the top of any general-purpose ranking. The field-level  $h_2$  is the unit that should be the default for comparison, not the aggregate.

On the methodological side, Egghe’s power-law functional form is empirically well-supported: institution count alone explains 91–92% of the variance in  $h_2$  and  $h_3$  respectively. However, the theoretical compounding prediction  $\alpha_0\alpha_1 \approx \beta_1$  fails by 177%. The model is empirically useful; the underlying theory needs a different reading. External validation against the CWTS Leiden Ranking Open Edition 2024 (van Eck, 2024) confirms that both  $h_2$  and  $\varepsilon_2$  correlate strongly with the Leiden mean normalised citation score across 1,495 matched institutions, demonstrating that the efficiency normalisation does not erase the quality signal and that the metrics capture genuine differences in research excellence rather than differences in institutional size (Subsections 2.11, 3.8).

Several limitations constrain interpretation. Author disambiguation is the most consequential open problem: the four outlier cases identified in cross-validation (notably Lei Jiang,  $h_1^{\text{OpenAlex}} = 215$  against Web of Science = 115) confirm that merger errors cluster in non-Anglophone and common-name strata, exactly the authors whose inflated  $h_1$  values would propagate most damagingly into institutional and country-level rankings. That cross-validation sample was deliberately adversarial, however, and cannot by itself say how much of the full 17.1-million-author population is affected; a Monte Carlo sensitivity analysis run at population-wide error rates from 1% to a deliberately implausible 80% (Subsections 2.9, 3.11) finds the risk bounded rather than fatal, with institution- and country-level rank correlations declining only modestly before plateauing ( $\rho \geq 0.988$ ) rather than degrading toward noise at any rate tested, and the paper’s country-comparison claims holding throughout. The field-leadership claims are noisier and should be read with correspondingly more caution than the paper’s other claims. Arts and Humanities are systematically under-covered in all major citation databases: both threshold authors sampled at Oxford returned no value in Web of Science, Scopus, or Google Scholar, and h-index comparisons in book-based fields should be read with correspondingly less confidence. Field-specific rankings are also sensitive to how field membership is defined: a multi-field assignment (authors counted in every field comprising  $\geq 33\%$  of their output) yields highly stable overall rankings ( $\rho = 0.963$ ) but shifts Wageningen’s Environmental Science lead to the University of Colorado Boulder, confirming that the field-leadership claims should be read as reflecting depth of specialisation rather than unconditional supremacy (Subsection 3.4). OpenAlex’s inconsistent retraction flagging leaves a small, unquantified fraction of  $h_1$  values potentially inflated. Saudi Arabia’s efficiency rank (4th,  $\varepsilon_3 = 6.69$ ) is plausibly an artifact of affiliation-listing for funding rather than resident research presence (Bhattacharjee, 2011), a data-quality distortion visible in the output but not removable without ground-truth affiliation data. Finally,  $h_2$  is not comparable across fields of different size: Decision Sciences (max  $h_2 = 22$ ) cannot be placed on the same scale as Medicine (max  $h_2 = 119$ ), and the results should be read within fields only.

The most impactful near-term improvement to this framework would require a data source that does not yet exist: departmental affiliation in OpenAlex. If OpenAlex were to record the department within an institution alongside the institution itself, the hierarchy would extend naturally to researcher  $\rightarrow$  department  $\rightarrow$  institution  $\rightarrow$  country, with successive indices  $h_1$  through  $h_4$ . Departments are field-specialized by construction, so the step of inferring each author’s modal field from topic weights may

become unnecessary at the departmental level. I flag this as the single most valuable addition OpenAlex could make for scientometric work of this kind.

## Statements and Declarations

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